

Jayesh Patel

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PROFESSIONAL SUMMARY

Versatile Full-Stack & Systems Engineer with a B.Tech in Computer Science and deep expertise spanning systems-level programming, scalable backend architectures, high-performance computing, and cross-runtime integration. Proven track record designing and publishing production-grade open-source libraries, building high-throughput distributed APIs, and architecting data-intensive and intelligent software solutions. Strong foundation in software engineering fundamentals, dedicated to building performant, secure, and resilient systems from low-level runtimes to modern cloud and web platforms.

TECHNICAL SKILLS

Languages: C, C++, Rust, Python, PHP, JavaScript, TypeScript, Dart, Go, Java
Compilers & Systems: AST Parsers, FFI, Shared Libraries
Frontend: React, Next.js, Vue.js, AngularJS, Tailwind CSS, WebGL
Backend: Laravel, Node.js, Express.js, FastAPI, Flask, Django
Databases: PostgreSQL, MySQL, SQLite, MongoDB
Mobile: Flutter, Dart, Android (Java)
Cloud & DevOps: Linux, Nginx, GitLab CI, Render, Vercel
AI & Data Science: Deep Learning (PyTorch, TensorFlow), NLP, Computer Vision, Scikit-learn
AI Media & Generative: Stable Diffusion, GFGAN, RealESRGAN, CodeFormer, UVR
Design: Figma, Adobe Creative Cloud

WORK EXPERIENCE

Shreesofttech Dec 2024 – Present
Laravel Developer Hybrid

- Engineered production web architectures with Laravel, optimizing complex database workflows, Eloquent ORM queries, and RESTful API endpoints.
- Implemented secure backend services, database migrations, and caching layers to ensure high availability and robust performance.

WRTeam May 2024 – June 2024
PHP Developer Intern Remote

- Engineered and tested PHP web applications, improving codebase efficiency by 50%+ through dynamic fetching optimizations that significantly reduced file count and code volume.
- Built secure API endpoints and collaborated on cross-functional agile teams to deliver commercial client solutions.

CodSoft Jun 2024 – Jul 2024
Flutter Developer Intern Remote

- Built 4 cross-platform mobile applications with Flutter and Dart, implementing responsive UI components, state management, and local SQLite data persistence.
- Conducted code reviews and optimized app performance across Android and iOS targets.

CodSoft May 2024 – Jun 2024
Python Programming Intern Remote

- Constructed algorithmic automation suites, object-oriented software architectures, and backend scripting solutions across 5 Python applications spanning user management, game logic, security, and task management.

CodSoft May 2024 – Jun 2024
Data Science Intern Remote

- Developed machine learning pipelines, predictive statistical modeling, and exploratory data analysis with Pandas, NumPy, and Scikit-learn across 5 datasets (55+ visualizations), achieving 85%+ model accuracy.

PROJECTS

Panchang Core & JPL Ephemeris Ecosystem

[GitHub \(Core\)](#) | [GitHub \(C Engine\)](#)

C, CALCEPH, JPL DE440, PHP ext-ffi, Python ctypes, Dart FFI

- Architected an authentic Vedic Panchanga calculation engine powered by an independent C ephemeris engine (204 public functions, 462 constants) implementing JPL DE405/DE440, CALCEPH, Moshier, VSOP87, and ELP2000 theories.
- Computed Tithi, Nakshatra, Yoga, Karana, and 30 Muhūrtas with 0.001 arcsecond precision, orchestrating 336 unique festival identities and 126 vrat identities across Amanta and Purnimanta calendars.
- Distributed zero-overhead FFI bindings and precompiled native binaries across [PHP](#) (ext-ffi), [Python](#) (ctypes), and [Dart/Flutter](#) (dart:ffi).

CSSForge — Lossless CSS AST Refactoring Engine

[GitHub](#)

Rust (Edition 2024), Ratatui TUI, AST Parser, Crates.io

- Engineered a safety-first, lossless semantic CSS refactoring engine and interactive terminal workbench in Rust (Edition 2024), published on Crates.io.
- Implemented 28 AST transformation rules modernizing flat legacy CSS into native nesting, Range media queries, @layer consolidation, and :is() factoring with zero declaration loss.
- Supported raw stylesheets and embedded style blocks within Blade, Vue, Svelte, Astro, Twig, ERB, and HTML templates.

Lipimala — Deterministic Indic Script & Vedic Transliteration

[GitHub](#)

Dart, Node.js, Python, PHP, Unicode Vedic Tags

- Built a deterministic Indic transliteration suite published synchronously across 4 package registries: pub.dev (Dart), npm (JavaScript/TypeScript), PyPI (Python), and Packagist (PHP).
- Solved Brahmic many-to-one collapse via structured TransliterationResult envelopes and checksummed LIT1: Unicode-Tag metadata trailers.
- Enabled direct Devanagari ↔ Gujarati conversion with first-class preservation of Vedic svara markers (Udatta, Anudatta, Svarita).

Hindu Scriptures: Digital Repository & AI Scholar

[GitHub \(Web\)](#) | [GitHub \(Android\)](#)

Node.js, Express 5, Google Gemini AI, JavaScript, Java (Android APK)

- Developed a Sanātana Dharma digital repository indexing sacred literature across the Vedas, Upanishads, Bhagavad Gita, and Puranas with verse-level lexical search and Sanskrit root morphology.
- Integrated contextual AI scripture analysis using Google Gemini and delivered both a responsive web platform and a native Android application.

Laravel Gemini AI Translation Extractor

[GitHub](#)

PHP 8.3+, Laravel 11–13, Gemini AI, Fork (pcntl), Symfony Process, C kernel32 FFI

- Built an enterprise Laravel translation engine with dual-platform concurrency: Unix pcntl fork workers and Windows parallel Symfony Process workers, featuring smart token chunking, automatic rate-limit backoff, and atomic file writes.
- Engineered a native Windows interactive TUI via C kernel32.dll FFI console hooks providing arrow-key navigation and multi-select prompts matching Laravel Prompts, alongside a web-based Translation Manager with script-fault detection across a 249-language catalog.

Poetry Analyzer — Deep Linguistic Analysis Engine

[GitHub](#)

Python, FastAPI, PyTorch, Transformers, spaCy, Stanza, HuggingFace, Alembic

- Designed an async FastAPI backend orchestrating 1.5 GB+ HuggingFace Transformer models, spaCy, Stanza, and Indic NLP to compute poetic meter, rhyme patterns, phonology, and semantic motifs with sub-second latency.
- Implemented SQLAlchemy ORM connection pooling for SQLite, PostgreSQL, and MySQL, with Alembic schema migrations.

SQL to Laravel Migration Compiler

[GitHub](#)

JavaScript, Web Workers, Topological Sort, Laravel 11/12/13

- Built a privacy-first, client-side compiler transforming raw SQL database dumps into idiomatic Laravel 11–13 migrations and seeders with topological foreign-key dependency resolution, vector/spatial type mapping, and environment-aware dialect generation across MySQL, PostgreSQL, MariaDB, and SQLite.

EDUCATION

Parul University
B.Tech in Computer Science and Engineering
CGPA: 7.29

Vadodara, Gujarat, India
2021 – 2025

CERTIFICATIONS

- [PHP Developer](#)
- [Python Programming](#)
- [Data Science](#)
- [Flutter Developer](#)
- [Data Analytics with Python](#)
- [Laravel](#)

SOFT SKILLS

Analytical Thinking, Problem Solving, Systems Architecture, Attention to Detail, Technical Research, Time Management, Adaptability, Communication, Teamwork, Leadership